

Features Implemented

- Home page, login page, and sign-up page
- Account creation with name, email, password and role selection: Student or Admin
- Client-side validation with error messages
- Automatic redirecting to the dashboard page for already logged-in users
- Student dashboard with summary statistics like enrolled courses, overall completion progress and grade average
- Ability to enroll in available courses directly from the dashboard
- Navigation to each course's detail page from the dashboard
- Interactive calendar showing upcoming assessment due dates
- Day panel listing all assessments scheduled for a selected date
- Course info page displaying course details, current weighted average with a progress bar, number of pending assessments and a course description with the course information's
- Navigation from the course info page to grades and assessments for that course
- Fixed drop course button on the course info page
- Grades page with a per-assessment breakdown of earned marks, weight along with the overall course average
- Assessments page listing all visible assessments across enrolled courses with status such as Pending or Submitted
- File submission and resubmission via a modal with file type and size validation
- Admin course management page to create/edit, enable/disable, and delete courses
- Admin assessment management page to add, edit, and delete assessments per course
- Admin grades management page to select a course and student, enter or edit earned marks and feedback
- Admin statistics page with a graded vs. pending pie chart and a per-course average bar chart

Backend

For the backend, it is built with Node.js and Express which is connected to MongoDB via Mongoose. Authentication is handled with JWT where the registration/ login routes validate inputs, hash passwords alongside returning tokens. All protected routes verify the token with middleware and admin-only routes apply an additional role check to confirm the validity.

Course routes allow the students using the website to access their enrolled courses/all available ones, enroll in a course, and drop a course. Admins can create, update, edit and delete their courses. A server-side weighted average endpoint calculates each student's grade using their submission results. Additionally, the assessment routes let the admins create and manage the assessments while students can only retrieve the visible ones. There's a visible completion status and earned marks attached from the student's submissions. Submission routes handle the uploaded files via Multer and flag late submissions automatically. Furthermore, it allows admins to grade students and download their files.